

Use Case LM-1.1.2: Initialize Capsule

Scope: Launch Mission

Primary Actor: Flight Director

Secondary Actors: None

Related Use Cases: *Use Case 1: Initialize Launch*

Stake Holders & Interests:

- Flight Director: Wants to set up a Launch via initializing all the Launch Parameters, System and Sub-Systems Including:
 - Capsule/Payload
- Mission Planner: Wants the Mission Initialized with the appropriate Mission Parameters loaded

Pre-Conditions: The Mission System is running and ready to initialize and monitor the Capsule/Payload.

Post-Conditions: The Capsule is initialized with the appropriate parameters.

Flow of Events:

Flight Director	System
1. Mission Parameters file (See <i>Use Case 1 Initialize Launch</i>)	
	2. Finds the Capsule Parameters File and location
	3. Goes to the file path of the Capsule Parameters File
	4. Opens the Capsule Parameters File
	5. Reads the Capsule Parameters File
	6. Initializes the Capsule Data with the appropriate Capsule Data read in from the Capsule Parameter File
	7. Initialization Status returned to the Flight Director

Extensions (or Alternative Flows):

- 2a,3a If the System cannot find the location of the Capsule Parameters File, then the System returns an Error to the Flight Director: indicating not a legitimate file path. The Flight Director determines the next course of action.
- 4a If the System cannot open the Capsule Parameters File, then the System returns an Error to the Flight Director: indicating unable to open the Capsule Parameters File. The Flight Director determines the next course of action.
- 5a If the System cannot read the Capsule Parameters File, then the System returns an Error to the Flight Director: indicating unable to read the

Capsule Parameters File. The Flight Director determines the next course of action.

Notes/Constraints:

- *a This Use Case documents the Capsule Initialization, not the Capsule Monitoring (See *Use Case 4.1: Monitor Initialization*) for Capsule Monitoring
- *b The next course of action for the Flight Director is aborting the mission or choosing another Mission Parameter File
- 6a The System indirectly alerts the Stake Holders via an upward cascade of alerts/queries (See *Use Case 1.1 Initialize Rocket*)

Special Requirements:

- None

Technology & Variations List:

- There could be many types for Mission Parameters Files in different formats for the components. The System will need to account for all the different types of files.
- Regardless of file type, the System Data should follow a standard.

Frequency of Occurrence:

- Everytime a Mission is planned or simulated.

Open Issues:

- None